Technical Report: Diabetes 30-Day Readmission Prediction

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Project Overview

Predicting hospital readmissions using machine learning to enable proactive healthcare intervention and reduce avoidable costs.

Executive Summary

This report describes a machine learning approach to predicting 30-day hospital readmission for patients with diabetes using multi-hospital encounter data from 1999–2008. After exploratory analysis and data preparation, supervised learning classification models were trained to identify patients at higher risk of readmission. Results indicate that prior healthcare utilisation and patient complexity (e.g., inpatient visits, number of diagnoses, medication burden, and length of stay) are among the strongest predictors. The report concludes with actionable recommendations for targeting follow-up care and outlines limitations and future enhancements.

1. Problem Statement

Hospital readmissions, particularly within 30 days of discharge, are a critical indicator of healthcare quality and cost efficiency. Diabetic patients are at a significantly higher risk of readmission due to the chronic and complex nature of the disease.

The objective of this project is to:

- Identify key factors contributing to 30-day hospital readmissions among diabetic patients
- Develop a predictive model to classify patients based on their likelihood of readmission
- Support healthcare providers in making data-driven decisions for proactive patient management

Ultimately, the goal is to improve patient outcomes while reducing unnecessary healthcare costs.

2. Data Description

The dataset consists of patient records collected from multiple hospitals over a 10-year period (1999–2008). It contains both clinical and administrative data.

2.1 Key Feature Categories

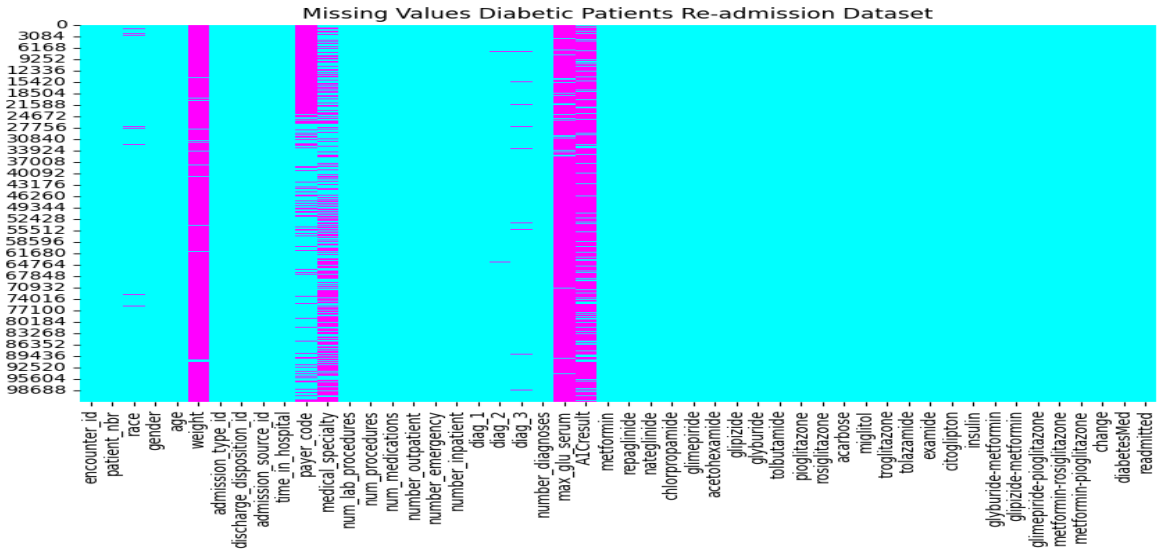
- **Patient Demographics:** age, gender, race
- **Hospitalization details:** time_in_hospital, admission type, discharge disposition
- **Medical history:** number_diagnoses, primary/secondary diagnoses
- **Healthcare utilization:** number_outpatient, number_emergency, number_inpatient
- **Medications:** insulin, metformin, and other diabetes-related drugs
- **Target variable:**
    - readmitted → indicates whether a patient was readmitted within 30 days

3. Methodology

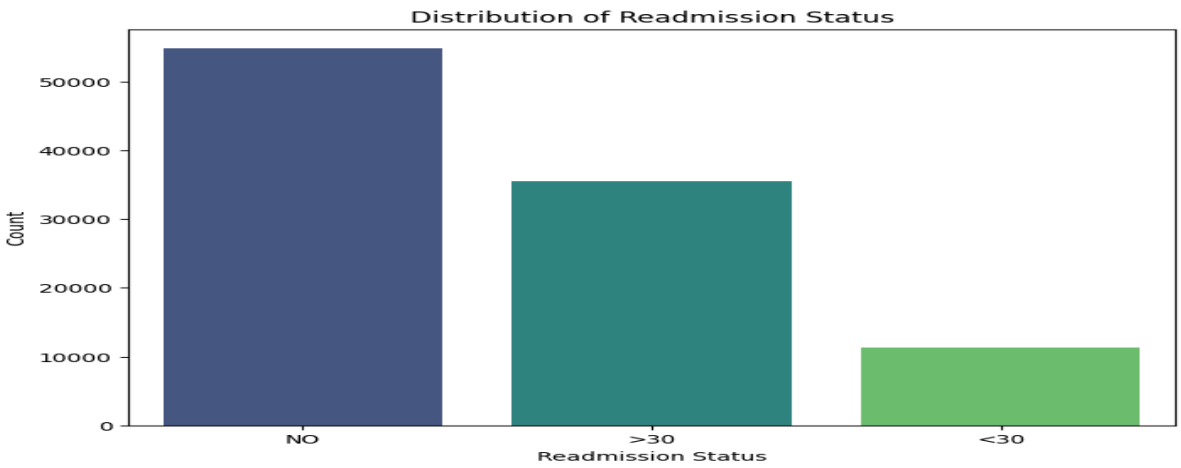
3.1 Exploratory Data Analysis (EDA)

EDA was conducted to understand the structure, distribution, and relationships within the dataset.

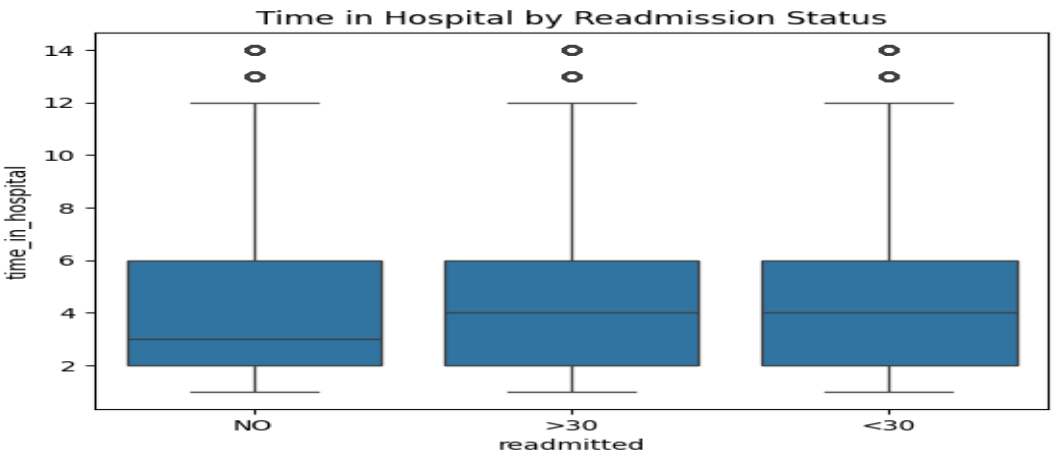
Key steps included:



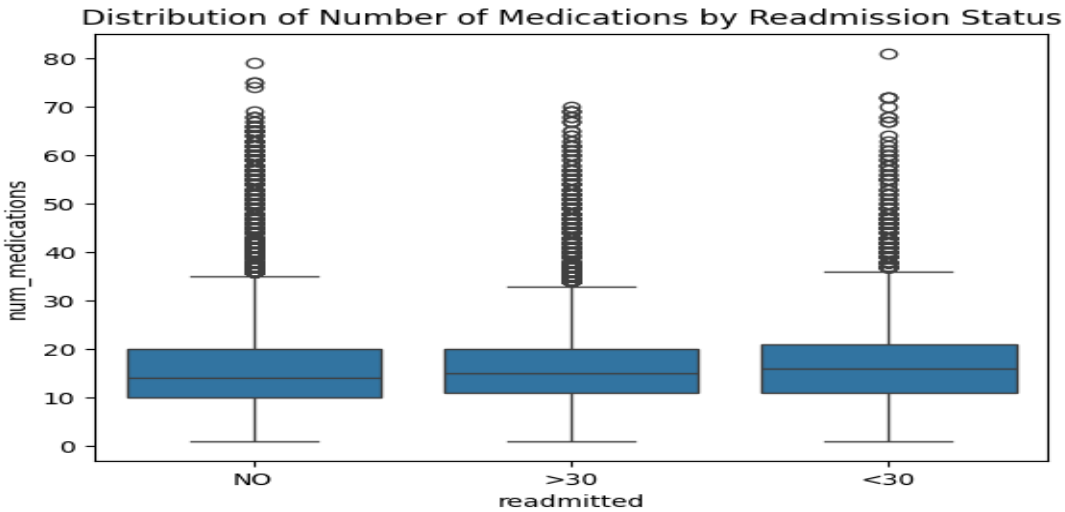
- **Figure 1:** plot showing missing values in the dataset



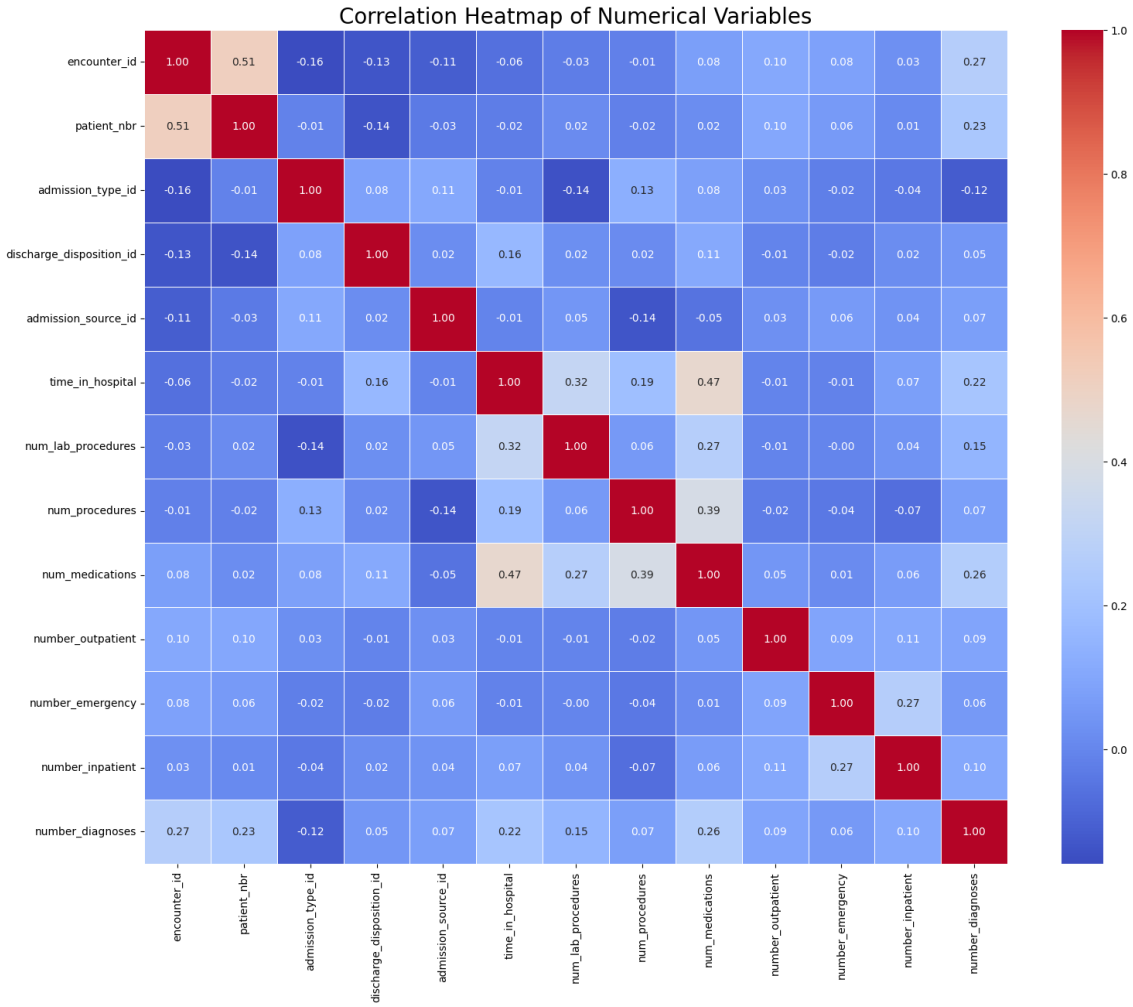
- **Figure 2:** Bar Chart showing Readmission Distribution



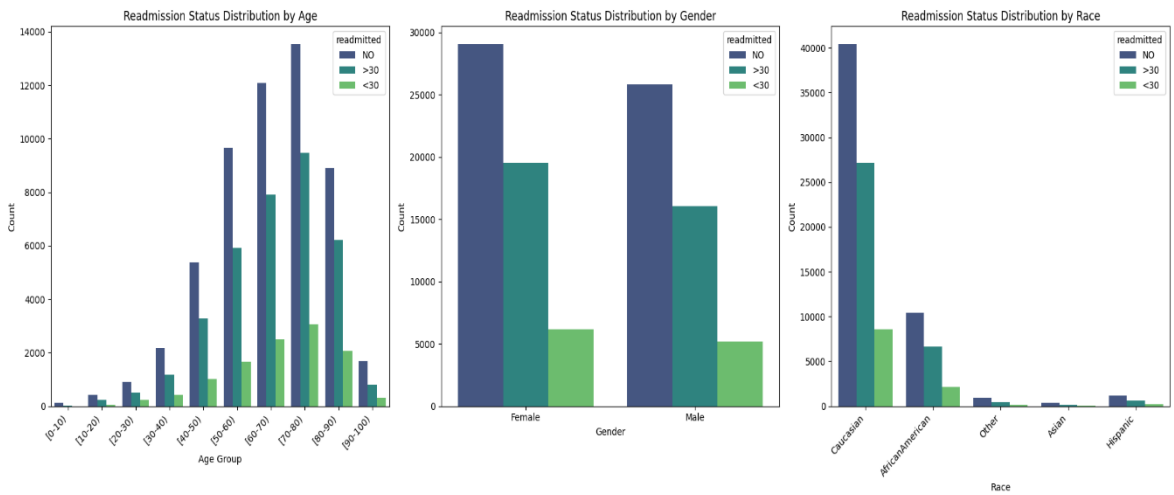
- **Figure 3**: Analysing distributions of numerical features (time in hospital)



- **Figure 4**: Analysing distributions of numerical features (number of medications)



- **Figure 5**: Correlation Heatmap of Features



- **Figure 6:** Visualizing trends using bar charts and distribution plots

Key Observations

- Readmission is moderately imbalanced
- Patients with higher hospital visits and diagnoses tend to show higher readmission rates
- Medication usage varies significantly across patients

3.2 Data Cleaning and Preprocessing

To ensure data quality and model readiness:

- Missing values were handled:
  - ✓ Columns with excessive missing values were dropped
  - ✓ Remaining missing values were imputed (e.g., mode for categorical variables like race)
- Data inconsistencies:
  - ✓ Replaced invalid entries (e.g., "?") with null values
- Categorical encoding:
  - ✓ Converted categorical variables into numerical format using encoding techniques
- Target transformation:
  - ✓ Converted readmitted into binary classification (e.g., readmitted vs not readmitted)

3.3 Feature Engineering

To improve model performance and interpretability:

- Created aggregated features:
  - ✓ total_visits = outpatient + emergency + inpatient visits
- Simplified medication features:
  - ✓ Converted multiple medication columns into binary indicators (used vs not used)
  - ✓ Created summary features such as medication count
- Selected relevant features based on domain knowledge and EDA insights

3.4 Modelling

Multiple machine learning models were explored, with a focus on classification.

- Models used:
  - ✓ KNN Classifier
  - ✓ Decision Tree
  - ✓ Random Forest Classifier (primary model)
- Steps:
  - ✓ Train-test split
  - ✓ Model training on training data
  - ✓ Select the best model based on performance

4. Results and Evaluation

Model performance comparison:

Model	Accuracy	AUC-ROC	Explanation
K-Nearest Neighbors	91.0%	0.56	High accuracy but weaker at distinguishing classes
Decision Tree	90.4%	0.52	Simple but less powerful
Random Forest	91.0%	0.61	Best overall model

4.1 Key Findings

- The Random Forest model performed best at separating patients who will be readmitted from those who will not.
- The model demonstrated reasonable predictive performance
- It was effective at identifying high-risk patients, though some class imbalance challenges remained

5. Insights and Recommendations

5.1 Insights

- Patients with multiple diagnoses are more likely to be readmitted
- Frequent hospital visits strongly correlate with readmission risk
- Higher medication counts indicate more complex cases
- Longer hospital stays may signal severe conditions

5.2 Recommendations

- **Target High-Risk Patients:** Focus on patients with high visit frequency and multiple diagnoses
- **Post-Discharge Monitoring:** Implement follow-up programs for patients with long hospital stays
- **Medication Management:** Optimize treatment plans for patients on multiple medications
- **Preventive Care Strategies:** Reduce emergency visits through early intervention

6. Limitations and Future Work

6.1 Limitations

- Missing or incomplete data in some features
- Lack of temporal data (no exact admission/discharge timestamps)
- Class imbalance affecting model sensitivity
- Limited interpretability compared to advanced models

6.2 Future Work

- Apply advanced models (e.g., gradient boosting, XGBoost)
- Incorporate time-series analysis if timestamp data becomes available
- Deploy the model using a web application for real-time predictions
- Integrate additional patient data (e.g., lifestyle, socioeconomic factors)

7. Conclusion

This project successfully demonstrates how machine learning can be applied to predict hospital readmissions among diabetic patients. By identifying key risk factors and building a predictive model, the analysis provides actionable insights that can help healthcare providers improve patient outcomes and reduce costs.

References

- Tayal, S. Diabetic Patients Readmission Prediction [dataset]. Kaggle. Available at: https://www.kaggle.com/datasets/saurabhtayal/diabetic-patients-readmission-prediction (Accessed: 31 March 2026).